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The Design and Implementation of a Low-Cost Electroencephalogram to Predict Neural Disorders

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***Abstract*— We designed and developed an EEG to help predict neural disorders. This project achieved both high accuracy and low cost with cost-effective components such as the AD622ANZ instrumentation amplifier and TL048x operational amplifier. Signal processing software and hardware filters were implemented. In addition, a machine learning approach was utilized to develop a binary seizure classifier. Seizure noise was simulated, and future work would revolve around collecting live data from epileptic patients. Future work could show that our design could detect and diagnose other neural disorders. We aim to make this design a closed loop system and BCI (Brain-Computer-Interface) compatible.**

***Index Terms*— Electroencephalography, Epilepsy, Instrumental Amplifier, Machine Learning, Operational Amplifier, Signal Processing**

I. Introduction

Contribution Summary

- We designed and constructed a working low-cost electroencephalogram (EEG) using economical components in the form of a printed circuit board (PCB) with electrodes.
- We designed and tested signal processing software.
- We simulated artificial seizure noise to work around not having access to data from epileptic patients.
- We evaluated our implementation on a binary classification ML algorithms, and found that Extremely Random Trees had an Area Under Curve (AUC) score of 97.4% on test data.

A. Background Research on Epilepsy

Epilepsy is a disorder of the central nervous system (CNS), affecting about 1.2% (3.4 million people) in the US, and more than 65 million globally [1]. Additionally, about 1 in 26 people will develop epilepsy at some point during their lifetime [2]. There are numerous kinds of seizures, each with different symptoms such as losing consciousness, jerking movements, or confusion. Some are significantly harder to detect visually, and the patients will usually exhibit symptoms such as not responding or staring blankly for a brief period of time. These are known as absence seizures. Seizures can happen unexpectedly and can result in injuries such as falling, biting of the tongue, or driving accidents [3]. Hence, seizure detection is of utmost importance for patients under medical supervision that are suspected to be seizure prone.

B. Background Research on EEG Technologies

An EEG, or electroencephalogram, measures electrical activity in the brain and can be used in the diagnosis and treatment of head injuries, infections, brain tumors, epilepsy, sleep disorders, coma, and diseases such as Alzheimer's [4]. It can also be used to monitor a patient during surgery. An EEG's output is a time-series. To detect epileptic, trained medical professionals compare the output to a standard EEG signal. Although misinterpretation of medical data is a concern across many fields, EEG data is more commonly labeled incorrect. This may ultimately stem from the essence of EEG reading, a pattern recognition skill that cannot be acquired from textbooks, articles, supplements, or guidelines, a process far more enigmatic than appreciating a lesion on an MRI study. Various image processing, signal processing, and machine-learning based techniques are employed to analyze epilepsy, using spatial and temporal features. The nervous system that generates the EEG signal is considered nonlinear and the EEG signals exhibit chaotic behavior.

In addition, an EEG test is expensive, both for a patient and for the healthcare provider. For patients covered by health insurance, typical out-of-pocket costs for an EEG would consist of a copay or coinsurance of 10%-50%. For patients not covered by health insurance, an EEG typically costs $200-$700 or more for a standard EEG – or up to $3,000 or more if extended monitoring is required. For example, Elizabeth Regional Medical Center [5] in Nebraska, charges $650 for an awake-and-drowsy EEG and about $3000 for EEG monitoring. An EEG amplifier tends to be more complex than a regular amplifier because, as with all physiological data, it is very noisy. EEG signals at the scalps are in the order of microVolts, which requires amplification. Furthermore, noise from dead skin cells, scalp oils, hair, air gaps, all add to the electrical noise of the system, which is usually subdued by costly insulation. Electrodes themselves are often coated in gold to have the best coupling to detect neural signals. More electrodes lead to better, more consistent data acquisition, but drives up the cost significantly. Adding intellectual property costs and company overhead costs (driven up by certifications of medical agencies like the FDA) yields higher prices of EEG systems for hospitals, amounting up to $50,000. This does not account for annual maintenance.

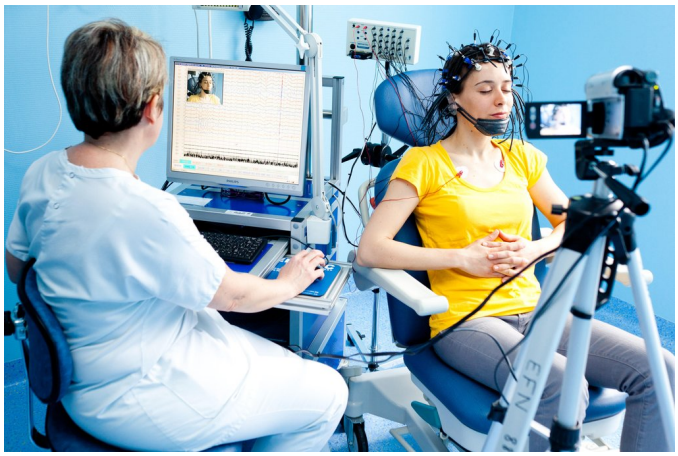

Fig. 1. A traditional EEG device at hospitals [6].

While companies such as NeuroSky, OpenBCI, BioSemi, and Cognionics attempted to create cheaper EEG technologies, their price range is $25,000+, well above the budget of hospitals in many developing countries [7]. Thus, this project aims to develop a cheaper but accurate neural signal acquisition system complemented with

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a ML algorithm that removes expert medical training out of the system. We acknowledge that this is no substitute for a EEG test with hospitals, but also want to provide a resource for those who cannot afford the test. Its use would be similar to that of other early indicator devices, such as a pulse oximeter [8].

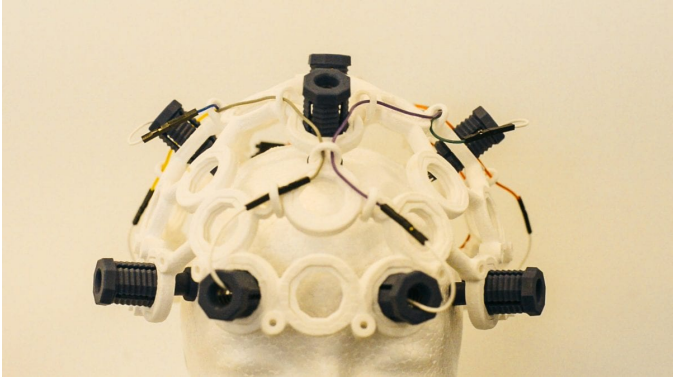

Fig. 2. OpenBCI Headset starting from $5,999.

C. Research Question

As EEGs are expensive and require a trained professional for operation, can we construct a low-cost circuit using digital integrated circuits (ICs), write signal processing software, with an ML architecture to classify ictal states?

D. Hypothesis

If a self-contained, low-cost EEG can be made with digital ICs with signal acquisition and processing software with a ML algorithm to classify seizures, it would provide access to epilepsy and other neural disorder diagnosis and assistance for the underprivileged and underfunded institutions.

II. Design

In this project, we utilized an Arduino Uno microcontroller [9], TDE-2143-C EEG Cup Electrodes [10], Electrode gel, Instrumental Amplifier AD622ANZ (x2) [11], Quad Operational Amplifier TL084x [12], ceramic capacitors (100nF, 220nF), resistors (560, 1.2K, 3.9K, 5.6K, 7K, 12K, 22K, 27K, 100K), wires, two 9V batteries, breadboard, multimeter, and an oscilloscope.

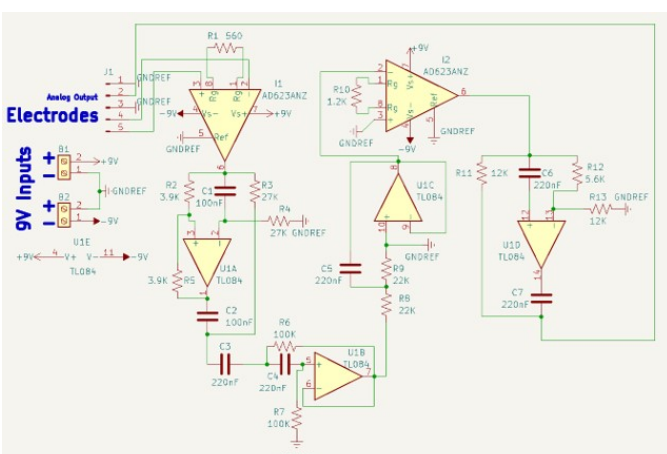

Fig. 3. Complete Schematic of Design

A. Circuit Overview

The hardware aspect of the design consists of 6 main sections: 2 Instrumental Amplifiers (gain = 91), Notch Filter (60 Hz, gain = 1), High Pass Filter (Fc = 7.2 Hz, gain = 1), Low Pass Filter (Fc = 32.9 Hz, gain = 1), and another Notch Filter (60 Hz, gain = 1). The circuit has two different grounds – the Arduino's ground is 0V while the circuit's ground is 3.3V. This was due to the voltage limitations of the Arduino digital IO ports [13]. The amplifiers are powered by the two 9V batteries. See Figure 3 for the schematic and Figure 4 for the completed prototype implemented on a breadboard.

Here are the list of hardware filters that we used in our circuit:

- 60 Hz Notch Filter, Gain = 1
- 7.2 Hz Cutoff High Pass Filter
- 32.9 Cutoff Low Pass Filter
- 2nd 60 Hz Notch Filter, Gain = 1

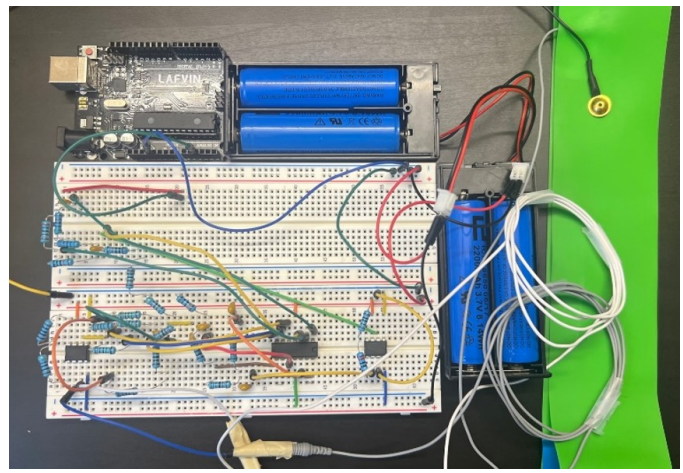

Fig. 4. Completed Prototype Design #1 on Breadboard. Electrodes on temporal nodes were taped to an elastic band to promote consistent contact during testing.

B. Instrumentation Amplifier

Neural signals are anywhere from 15 – 50 μV [14], so amplification is required to further process the signals and distinguish between brain waves and electrical noise. An instrumentation amplifier takes in two input voltages and outputs the difference between them, multiplied by a factor known as gain, which is given by

$$G = 1 + \frac{50.5\text{kOhm}}{\text{R}}. \tag{1}$$

where R is the resistance between pin 1 and 8 on the amplifier [15]. It is possible to construct an instrumentation amplifier using 3 operational amplifiers, which is a lower cost alternative. However, this method suffers from low common-mode rejection ratio (CMRR), so we chose to use an instrumental amplifier instead [16].

C. 1st Notch Filter (60 Hz, G = 1)

The largest source of electrical noise found in the circuit was due to power line interference. We noticed that it decreased when using a printed circuit board (PCB), but we chose to implement a notch filter into the circuit to remove most of the noise anyway. As described by Professor Ulaby, notch filters have a "severe reduction of gain around one particular frequency," and the 1st notch filter was used to filter out noise before the signal was subject to more gain [17]. The notch

frequency is determined by the following equation, where R = R3 = R4 = 27 kOhm in the circuit, and C = 100 nF:

$$f = \frac{1}{2\pi \mathrm{RC}}. \tag{2}$$

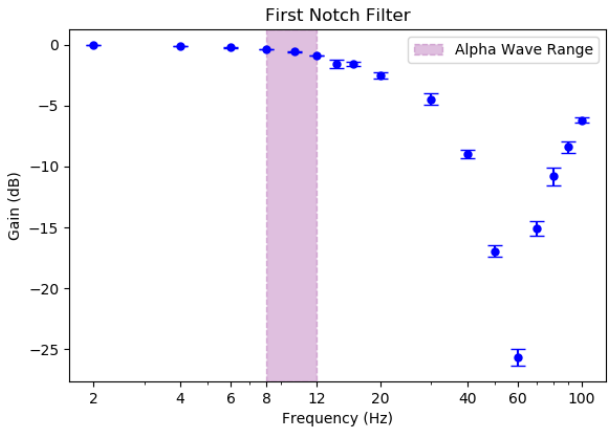

Fig. 5. 1st Notch Filter Response

D. High Pass Filter (Fc = 7.2 Hz)

The high pass filter passes every frequency above the cutoff frequency of 7.2 Hz. This was implemented to remove the noise generated from the galvanic skin frequencies (around 6 Hz). The 1.2 Hz difference is hypothesized to not affect the electron charge density at the tips of the electrodes. See Figure 6 for the filter response.

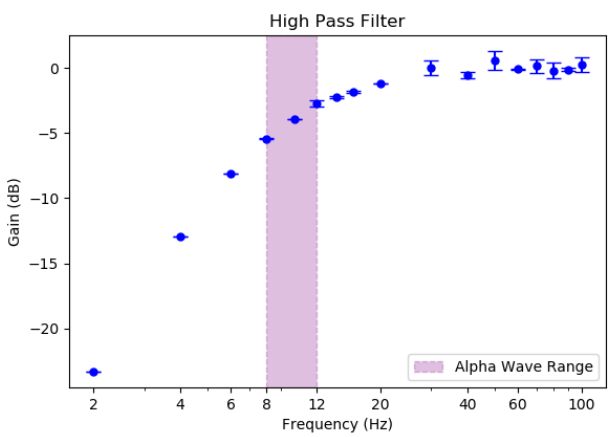

Fig. 6. High Pass Filter Response

E. Low Pass Filter (Fc = 32.9 Hz)

The EEG waves of interest to the project are alpha (8-12 Hz) and Beta (12-30 Hz), where major fluctuations occur during seizures [18]. Thus, we can safely remove frequencies above 30 Hz. A second order filter design was used. See Figure 7 for the filter response.

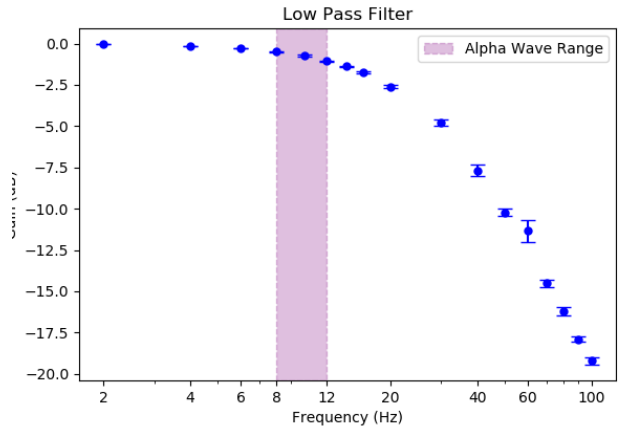

Fig. 7. Low Pass Filter Response

F. Instrumental Amplifier with Variable Gain

The 90-460 gain is cascaded with the 90x gain from the first instrumental amplifier. While alpha wave amplitudes vary from person to person, they usually range from 15 - 50 μV. Here, we assumed that the weighted average of the range would be 30 μV. This means that the end voltage could range from $909030 \cdot 10^{-6} = 0.243$V to $4609030 \cdot 10^{-6} = 1.242$V. Variable gain is achieved through the use of resistors in series and parallel, with the added 10 kOhm potentiometer, and

$$90 \leq G \leq 460 \tag{3}$$

G. 2nd Notch Filter (60 Hz, G = 1)

Another 60 Hz notch filter is necessary at the end of the circuit due to power line interference seeping through prior steps. This portion of the circuit is an exact copy of the 1st notch filter. See Figure 8 for the filter response.

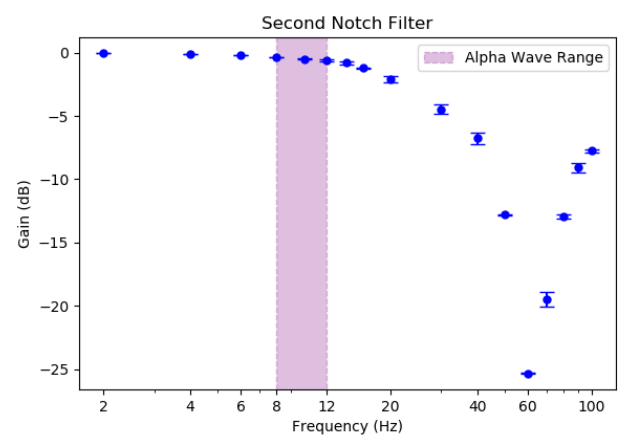

Fig. 8. 2nd Notch Filter Response

H. PCB

After building the initial prototype on the breadboard, we designed a custom printed circuit board (PCB) which was manufactured by commercial vendors such as JLCPCB. This was done to aid with

noise cancellation due to movement of wires. The total cost of the completed circuit, with electrodes and batteries, is $22.17. See Fig. 9 for the KiCad design, Fig 10. for PCB without soldered components, and Fig. 11 for final PCB with soldered components.

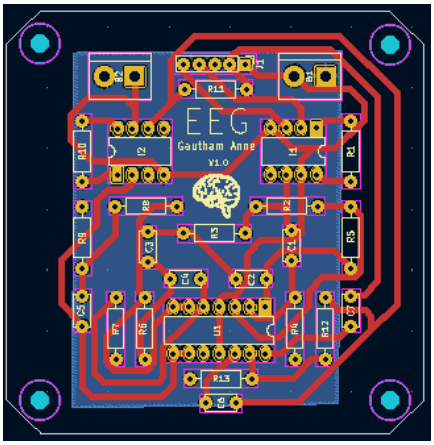

Fig. 9. PCB, as designed on KiCad.

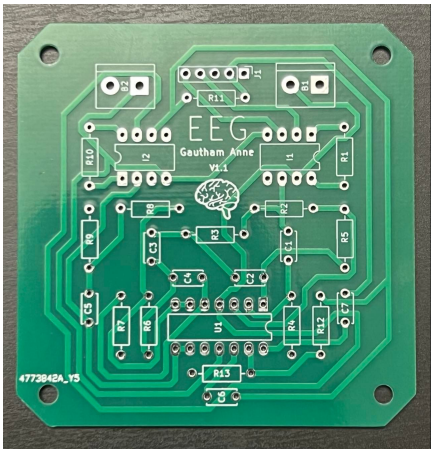

Fig. 10. Clean PCB.

I. Circuit Debugging Tips

We compiled the following tips to debug the circuit:

- Check individual sections of the circuit with multimeter and 5V input.
- Make sure that no component is overstrained or has excess current.
- Switch out Operational Amplifiers, as we noticed that these components are most likely to fail due to excess current.

If the circuit works, but you do not see brain waves:

- It is likely that the electrode placement is wrong. This happens in many commercial devices too, as neural signals are detectable at higher voltage only at specified locations on the scalp ([19], [20], [21], and [22]). Try asking someone to place the electrodes on you, wash your hair, receive a hair cut at the electrode placement locations, etc.
- We also recommend the use of electrode gel to at the contact site between skin and electrode to promote conductivity [23].
- Switch the electrode wires feeding into the instrumentation amplifiers, as you might be amplifying negative voltages. Make sure to use gold-plated electrodes, as gold protects the component from corrosion, heat, wear and helps ensure a reliable electrical connection [24].
- Switch out Operational Amplifiers.

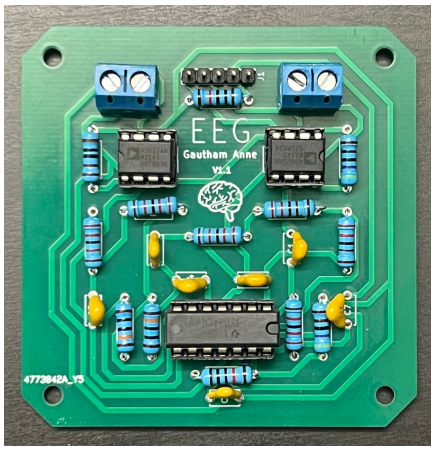

Fig. 11. Soldered PCB.

J. Electrode Placements

There are many possible electrode placements, depending on the focus of the study. For the purposes of this project, we use three electrodes: one above the left mastoid (the bone found behind the left ear), which aids with noise cancelling, one located at T7, and the other located at T8, as indicated by the extended Modified Combinatorial Nomenclature (MCN) 10-20 system for electrode placement [25]. The voltage difference oscillations between the T7 and T8 electrodes target brain waves from the temporal lobe. This choice was made since temporal lobe epilepsy is the most common form of epilepsy [26]. See Fig. 12 for the electrode placement according to the MCN 10-20 system.

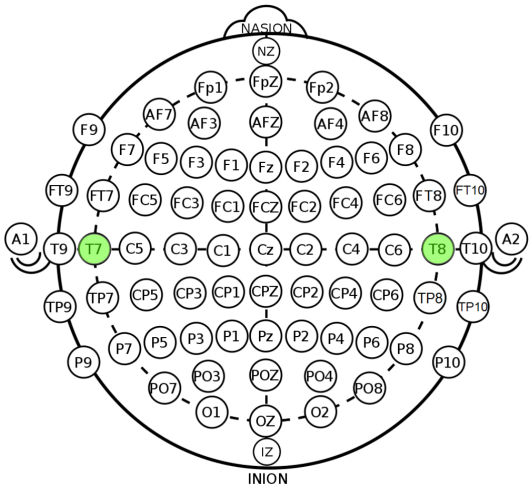

Fig. 12. Extended diagram of the MCN 10-20 system for placing electrodes. The chosen placements are highlighted in green. Ground electrode not shown.

K. Experimental setup

The human subject tapes three electrodes to their scalp at various descriptions as described in Section 4. Two of the electrodes are connected to an instrumentation amplifier, which aims to amplify the voltage difference oscillations between them. These oscillations are

the brain waves we target, and the circuit employs several filters to remove other electrical signals (noise). The third electrode serves as a 'ground' electrode to remove noise resulting from the electrode-skin contact. The voltage output is measured by the Arduino's digital IO port, which is then received by MATLAB's Simulink setup, enabling the user to apply various signal processing methods to further remove noise. To adjust the potentiometer to account for differences in alpha wave readings between different people, take readings while making sure that the person is stationary and at rest (calm state). Make sure the voltages do not fluctuate off screen, but avoid making the scale too small, since errors incurred in Arduino digital readings would be increased in proportion.

Finally, after verifying that the EEG signals are being captured, the subject runs the python script outlined in the 'Algorithm' Section. After further signal processing by various high/low pass filters, images of both the raw data and the processed signal is outputted.

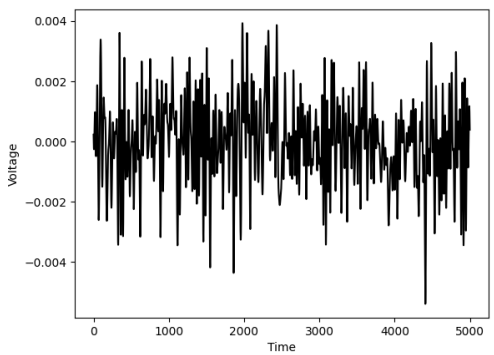

Fig. 13. Example of acquired signal from human subject.

III. Seizure Simulation

Noting the assumptions of previous research studies ([27], [28], [29], and [30]) we modeled seizures by overlaying random frequencies over a standard EEG signal. Thus, we wrote a python script that added 30 sine waves of frequencies from 8-12 Hz (randomly chosen), with an amplitude of an FIR (Finite Impulse Response) filter of the amplitudes of the standard signal as suggested by [31] and [32]. Phase shifts from 0 to 2π were assigned randomly to encourage constructive and destructive interference, in an attempt to increase the randomness of the generated signal. This models ictal states of epileptic patients. See Fig. 14 for samples of standard and ictal signals.

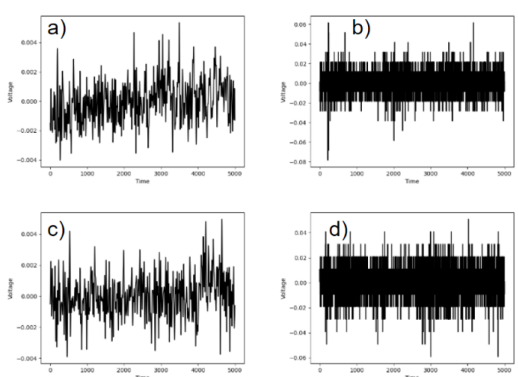

Fig. 14. Samples of Standard (a, c) vs Modeled Ictal (b, d) signals.

IV. Binary Classification Algorithms

To create the various ML algorithms to classify whether the signal is in an ictal state or not, we utilized the Sklearn python library, which provides a wide range of unsupervised and supervised ML algorithms. We implemented all of the following algorithms and compared their accuracy using the AUC (Area Under Curve) as our figure of merit (FOM). The higher the AUC, the better, ranging from 0 to 1 ([33] and [34]). We used AUC as our FOM because it is both scale-invariant and classification-threshold invariant [35].

A. K-Nearest Neighbors (KNN)

KNN stands for K-Nearest Neighbors, and it is a classification model that classifies the sample based on the class of k samples that are closest to it. For example, if k=3, and all three of the samples nearest the sample in question is class 1, then the sample in question would be classified as class 1. If two out of the three nearest the sample in question is class 1, then the sample in question would have a probability of 66% to be classified as class 1.

B. Logistic Regression

Logistic Regression is a generalized linear model, which takes input features and performs a linear function on them by projecting sample data onto the line. The linear function is performed by summing the log of the likelihood of each sample point to produce a best-fitting logistic regression by maximizing the log of the likelihood, which is the same as maximizing the likelihood [**?**]. In a binary classification model, the best fitting function would predict whether the state is ictal or not close to 100% of the time.

However, maximizing the cost function leads to overfitting the function, so we penalized it by performing L1 and L2, or elastic net regularization [37]. Lastly, we used a gradient descent algorithm to minimize the cost function with regularization. The model returns the probability of a state, so we have to set a threshold value to classify it as a positive class.

C. Stochastic Gradient Descent

SKlearn's SGDC estimator implements regularized linear models with stochastic gradient descent (SGD) learning: the gradient of the loss is estimated each sample at a time and the model is updated along the way with a decreasing learning rate [38].

D. Naive Bayes

The Naive Bayes Classifier assumes that all the features of the signal are not related to each other, such that the probability of all the features occurring are the product of the probabilities of each feature occurring. That is, the probability of the features occurring is independent from one another.

E. Decision Tree Classifier

A decision tree asks a question, and then classifies the sample based on the answer. The classification can be into categories or numeric. The algorithm works by repetitively separating data into sub regions of the same class and the tree ends when the algorithm has divided all samples into classes that are pure, or by meeting some criteria of the classifier attributes.

F. Random Forest

We took inspiration from previous related studies, for which a majority utilized random forests to classify the state of a neural signal. We built the Random Forest by bagging decision trees that are not correlated with each other by bootstrapping the set of samples or by using a random number of features at each split. RF combines the simplicity of decision trees with flexibility, resulting in a vast improvement in accuracy, which is one of the weaknesses of decision trees as a classifier.

G. Other Algorithms

In addition to the algorithms we discussed in this section, we also implemented the following:

- Gradient Boosting Classifier
- Extremely Random Trees
- XGBoost Classifier

V. Results

A. Filter Responses

In the design section sections, the responses for the various filters were shown. Fig. 15 shows the total filter responses for the entire circuit. The filter gains were maximized at the 8-12 Hz band, which showcases the success of the circuit in amplifying the desired signals.

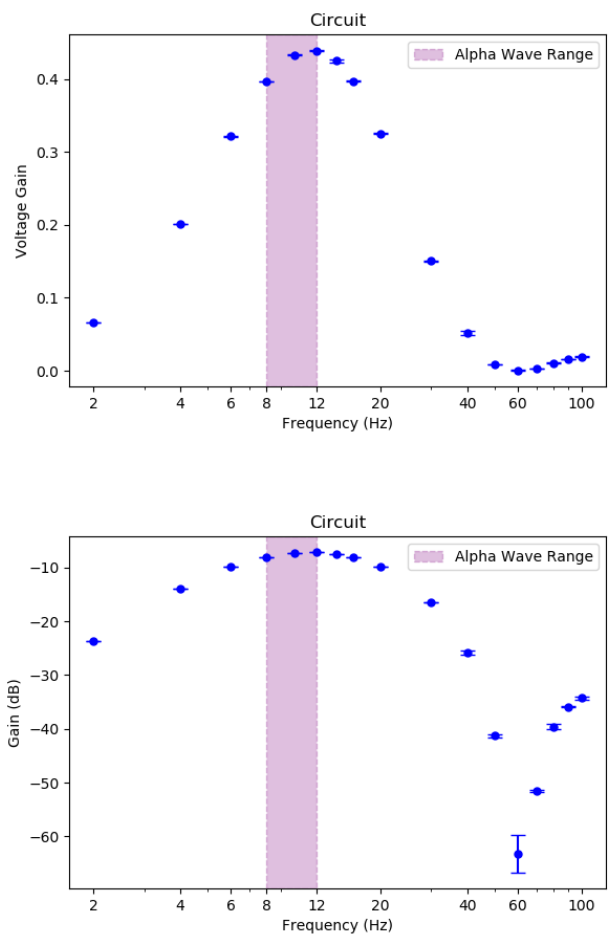

Fig. 15. Total Filter Responses

B. EKG Signal

After confirming our signal acquisition and processing was working perfectly, we first tested whether the EEG can acquire a heartbeat signal, akin to the EKG. This is because an EKG is an EEG with less sophisticated hardware and filtering processes. We were able to perfectly identify heartbeats in all of our 5 subjects. See Fig. 16 for an example of heartbeat signal we collected.

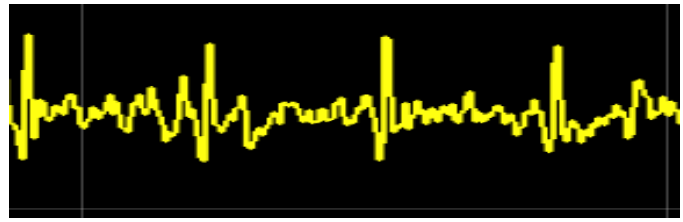

Fig. 16. Heartbeat signal of 17 year old subject shown. Processed through Matlab. Fourier transform of signal had peak at 1.2 Hz, equating to approximately 72 bpm, their resting heart rate [39].

C. EEG Signal

After acquiring an EKG signal, we used methods described in 2.K to acquire and process EEG signals. Fig. 13 shows an output of the signal after processing, and its Fourier Transforms reveals that the majority of the frequencies of the signal fall under the 8-12 Hz frequency band. Thus, our EEG was able to capture live neural signals.

D. Classifier Results

After training and validating the model based on the data set of collected standard EEG signals and modeled ictal signals, we have found that the Extremely Random Trees model performed the best, with an AUC score of 99.6%. This model had a lift performance of 4.3, meaning that it was 4.3 times more likely to predict whether a seizure is ictal or not as compared to randomly guessing. In the test set, its AUC was 97.4%. See Fig. 17 for a comparison of various models based on AUC and the AUC curve of Extremely Random Tree algorithm.

VI. Discussion

Interestingly, most of the models, except Logistic Regression and Stochastic-Gradient Descent, performed extremely well on binary classification. Logistic Regression performing relatively worse, with a test data AUC = 56.6%, can mainly be explained by its assumption of linearity between amplitude and time. Stochastic-Gradient Descent performing relatively worse, with a test data AUC = 56.4%, can mainly be explained by its tendency to get stuck at a local minima.

We expected the voltage gain to peak at the middle of the 8-12 Hz band, but as shown in Fig. 15, there is a shift towards the end of the spectrum. This is due to hardware limitations, as it is unreasonable to expect every resistor, capacitor, and IC circuit to function ideally. Figure 6 and 7 show that the 8-12 Hz frequencies are not at uniform sensitivity. An ideal electrical filter rejects unwanted frequencies but should have uniform sensitivity for desired frequencies.

After the High Pass Filter ($F_c = 7.2$Hz), frequencies closer to 12 Hz had a higher gain (-2.5 dB) while frequencies at 8 Hz had a lower gain (-5 dB), where gain is the logarithmic ratio between the output and input voltage of the preamplifier. That is, although the hardware High Pass filter rejected frequencies below 7.2 Hz well, it did not uniformly accept frequencies higher than 7.2 Hz.

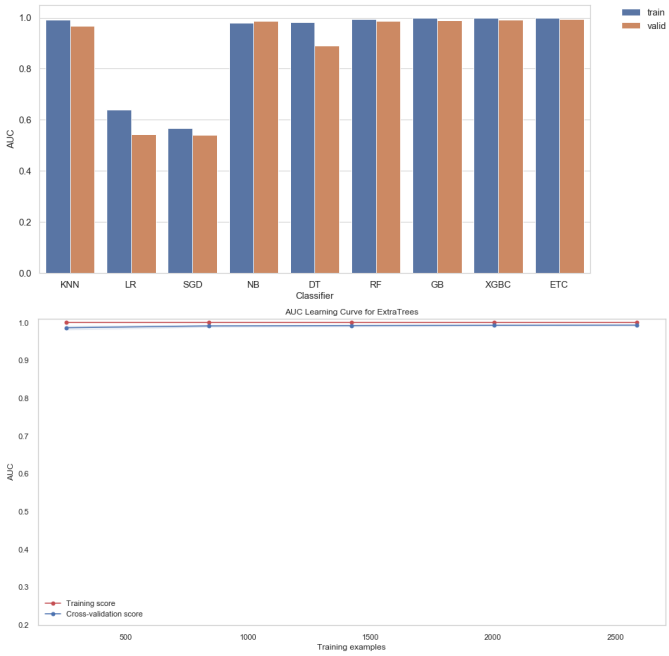

Fig. 17. Various models and their AUC scores on training and test data (top), and AUC curve with True Positive vs. False Positive (bottom) shown. Not much can be done to improve the prediction of the Extremely Random Tree algorithm, but future work could attempt to optimize hyperparameters.

Similarly, after the Low Pass Filter (F_c = 32.9Hz), frequencies closer to 12 Hz had a lower gain (-2.5 dB) while frequencies at 8 Hz had a higher gain (-0.1 dB). That is, although the hardware Low Pass filter rejected frequencies above 39.2 Hz well, it did not uniformly accept frequencies higher than 39.2 Hz.

Thus, our proposed solution is to incorporate software Chebyshev Type II FIlters convoluted with Cauer Filters to attempt to retain the sharp cutoff at unwanted frequencies but obtain uniform sensitivities for wanted ones. For example, hardware filters could be set at higher and lower cutoff frequencies (F_{LC} = 5.2 Hz and F_{HC} = 40 Hz), as to not rely on faulty components. This would necessarily be complemented with Chebyshev Filters at 8 and 12 Hz.

A. Independent Component Analysis (ICA)

The circuit has shown success in filtering out noise from a wide frequency range (caused by galvanic frequencies or power line interference). However, it is still subject to artifact signals unrelated to the brain waves of interest. In the future, we would like to experiment with independent component analysis (ICA). In related studies, ICA has proven to be a robust methods to separate brain waves and eliminate contamination from signals such as eye movement, blinks, muscle contractions, and EKG. ICA is a signal processing method to separate independent sources mixed together by a sensor. It recovers the original sources by multiplying the data with an unmixing matrix

$$U = WX$$

where X is the data with dimensions channel · time, U is the ICA source activities (component · time), and W is the ICA unmixing matrix. The technique attempts to find W by minimizing the Gaussianicity of each data set. To incorporate this, we assume the following:

- Mixing is linear at the electrodes.
- Delays are negligible.
- Component time courses are independent.
- The number of components are less than or equal to the number of channels.

B. Cost

The cost of our circuit is significantly less than current commercial options, at an impressive $22.17. However, to incorporate features such as ICA with more channels, it is inevitable that the price will rise. With bulk supply from various electronics distributors, such as DigiKey, it is possible to keep the total cost of the EEG system within an affordable range ($100 - $200). We would like to move away from the Arduino, or any microcontroller based on the atMega chip, because its analog to digital conversion is known to be very noisy. For example, we could utilize a RaspberryPi Microcontroller with an external ADC chip. In fact, if we choose to continue using the Arduino platform, we could utilize Phase-sensitive detection (PSD) to extract the voltage signal from external noise [40].s

C. Data Acquisition

Due to constraints of this project, we were unable to source EEG signals from patients with epilepsy, and had to simulate our own seizure noise. However, this presents a bias in our system, as we assume that seizures are random noise layered on top of a standard EEG signal.

VII. Conclusion

We developed a low-cost EEG ($22.17) that has signal acquisition and processing capabilities and implemented it on an Extremely Random Trees ML model, to which we found the AUC score on test data was 97.4% . Our project was very successful in acquiring neural signals, processing them, and predicting whether the patient is in a seizure state or not. Although accurate, it is not a replacement for clinical diagnosis, but should be taken in consideration when determining whether a patient has epilepsy. However, this project also revealed several hardware issues and ease-of-use issues that need to be further investigated with regards being a self-contained, accessible system.

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References

[1] Z. MM, K. R. National and state estimates of the numbers of adults and children with active epilepsy — United States, 2015. MMWR. 2017;66:821–825.

[2] Institute of Medicine (US). Committee on the Public Health Dimensions of the Epilepsies; England MJ, Liverman CT, Schultz AM, Strawbridge LM, Epilepsy across the spectrum: Promoting health and understanding. Washington: National Academies Press; 2012. s

[3] Hesdorffer, D C et al. “Estimating risk for developing epilepsy: a population-based study in Rochester, Minnesota.” Neurology vol. 76,1 (2011): 23-7. doi:10.1212/WNL.0b013e318204a36a

[4] Electroencephalogram (EEG)’ ’, HopkinsMedicine.Org, 2021, www.hopkinsmedicine.org/health/treatment-tests-and-therapies/electroencephalogram-eeg. Accessed 26 Nov 2022.

[5] Electroencephalogram (EEG), Nhs.U, 2017, www.nhs.uk/conditions/electroencephalogram. Acessed 26 Nov 2022.

[6] ”Saintelizabethonline.Com”. Saintelizabethonline.Com, 2022, www.saintelizabethonline.com/price-estimator-1.html. Accessed 26 Nov 2022.

[7] ”Biosemi EEG ECG EMG BSPM NEURO Amplifiers Systems”. Biosemi.Com, 2022, www.biosemi.com/faq/prices.htm. Accessed 27 Nov 2022.

[8] ”Pulse Oximetry”. Yale Medicine, 2022, https://www.yalemedicine.org/conditions/pulse-oximetry. Accessed 27 Nov 2022.

[9] ”Arduino - Home”. Arduino.Cc, 2022, www.arduino.cc/. Accessed 27 Nov 2022.

[10] ”Service Shop From GE Healthcare — GE Healthcare Support Services”. Services.Gehealthcare.Com, 2022, https://services.gehealthcare.com/gehcstorefront/. Accessed 27 Nov 2022.

[11] ”AD622ANZ”. Digikey.Com, 2022, Accessed 27 Nov 2022.

[12] ”TL084, Quad, 30-V, 3-MHz 13-V/μ s slew rate, In to V+, Jfet-input op amp”. Ti.Com, 2022, Acessed 27 Nov 2022.

[13] ”Arduino — Haseloff Lab:”. Haseloff Lab:, 2022, https://haseloff.plantsci.cam.ac.uk/education/biomaker/arduino/index.html. Accessed 27 Nov 2022.

[14] Başar, Erol. “Brain oscillations in neuropsychiatric disease.” Dialogues in clinical neuroscience vol. 15,3 (2013): 291-300. doi:10.31887/DCNS.2013.15.3/ebasar

[15] ”Instrumentation Amplifier Basics and Applications.” ElectronicsHub.Org, 2022, Accessed 27 Nov 2022.

[16] ”Instrumentation Amplifier Circuit Using Op-Amp.” Circuitdigest.Com, 2022, https://circuitdigest.com/electronic-circuits/instrumentation-amplifier-circuit-using-op-amp. Accessed 27 Nov 2022.

[17] F. U., “Frequency Response of Circuits and Filters,” in *Circuit Analysis and Design*, 1st ed. Michigan Publishing Services, Michigan, USA: 2018, pp. *530–533.*

[18] ”Understanding Brain Waves - Neurofeedback Alliance”. Neurofeedback Alliance, 2022, https://neurofeedbackalliance.org/understanding-brain-waves/. Accessed 27 Nov 2022.

[19] D. S., R. S., et. al, ”Consequences of EEG electrode position error on ultimate beamformer source reconstruction performance”, *Frontiers in Neuroscience*, 2014, DOI=10.3389/fnins.2014.00042, Accessed 27 Nov 2022.

[20] ”Troubleshooting With Your TENS Machine”. Babycaretens.Com, 2014, babycaretens.com/blog/troubleshooting-with-your-tens-machine-8951c4/. Accessed 27 Nov 2022.

[21] ”Troubleshooting Electrochemical Cell”. Redox.Me, 2022, redox.me/blogs/good-measurement-practices/troubleshooting-electrochemical-cell. Accessed 27 Nov 2022.

[22] ”Enercon Help Center”. Enercon.Force.Com, 2022, enercon.force.com/helpcenter/s/article/Minimizing-Electrode-Failures. Accessed 27 Nov 2022.

[23] ”NCI Dictionary Of Cancer Terms”. National Cancer Institute, 2022, www.cancer.gov/publications/dictionaries/cancer-terms/def/gel-electrode. Accessed 27 Nov 2022.

[24] V. M., V. M., M. R. et al. Gold-Plated Electrode with High Scratch Strength for Electrophysiological Recordings. Sci Rep 9, 2985 (2019). doi.org/10.1038/s41598-019-39138-w

[25] Sleep.Pitt.Edu, 2022, www.sleep.pitt.edu/wp-content/uploads/2020/03/10-20-system-el.pdf. Accessed 27 Nov 2022.

[26] ”Temporal Lobe Epilepsy (TLE)”. Epilepsy Foundation, 2022, www.epilepsy.com/what-is-epilepsy/syndromes/temporal-lobe-epilepsy

[27] G. M. et al. “EEG signal analysis using classification techniques: Logistic regression, artificial neural networks, support vector machines, and convolutional neural networks.” Heliyon vol. 7,6 e07258. 7 Jun. 2021, doi:10.1016/j.heliyon.2021.e07258

[28] G. C., A. P., N. M. et al. Automatic seizure detection based on imaged-EEG signals through fully convolutional networks. Sci Rep 10, 21833 (2020). doi.org/10.1038/s41598-020-78784-3

[29] G. B., R. D., K. B., Artificial EEG signal generated by a network of neurons with one and two dendrites, *Results in Physics*, Volume 20, 2021, 103699, ISSN 2211-3797, https://doi.org/10.1016/j.rinp.2020.103699.

[30] Arı, A., Analysis of EEG signal for seizure detection based on WPT. Electron. Lett., 56: 1381-1383. doi.org/10.1049/el.2020.2701

[31] Melinda, M., Syahrial, Yunidar, Al Bahri, and M. Irhamsyah. “Finite Impulse Response Filter for Electroencephalogram Waves Detection”. Green Intelligent Systems and Applications, vol. 2, no. 1, Apr. 2022, pp. 7-19, doi:10.53623/gisa.v2i1.65.

[32] D. N., U. S., et al., (2016). Comparing windowing methods on Finite Impulse Response (FIR) filter algorithm in Electroencephalography (EEG) data processing. 88. 558-567.

[33] B. A., ”AUC-ROC Curve In Machine Learning Explained”. Analytics Vidhya, 2020, www.analyticsvidhya.com/blog/2020/06/auc-roc-curve-machine-learning

[34] ”Understanding AUC - ROC Curve”. Medium, 2021, towardsdatascience.com/understanding-auc-roc-curve-68b2303cc9c5. Accessed 27 Nov 2022.

[35] ”Classification: ROC Curve And AUC — Machine Learning — Google Developers”. Google Developers, 2022, developers.google.com/machine-learning/crash-course/classification/roc-and-auc. Accessed 27 Nov 2022.

[36] ”1.2 - Maximum Likelihood Estimation — STAT 415”. Pennstate: Statistics Online Courses, 2022, online.stat.psu.edu/stat415/lesson/1/1.2. Accessed 27 Nov 2022.

[37] ”Elastic Net Regularization via Iterative Soft Thresholding”, web.mit.edu, web.mit.edu/lrosasco, Acessed 27 Nov 2022.

[38] ”Unsupervised Feature Learning And Deep Learning Tutorial”. Deeplearning.Stanford.Edu, 2022, deeplearning.stanford.edu/tutorial/supervised/OptimizationStochasticGradientDescent/. Accessed 27 Nov 2022.

[39] C. B., *Pattern Recognition and Machine Learning*. Springer Science+Business Media, LLC, New York, NY, 2006.

[40] K. S., Phase-Sensitive Detection in the undergraduate lab using a low-cost microcontroller, Arxiv, doi.org/10.48550/arXiv.1501.06074